

1899



This car from the 1920s advertises Bayer's drug bearing a Dutch slogan, which translates as "Overcomes all sufferings."

NEW ZEALAND-BORN PHYSICIST

ERNEST RUTHERFORD (1871–1937) was studying the radiation given off by uranium salts—first discovered by Henri Becquerel (see 1896). Rutherford was interested in the way radioactivity caused gases to be able to conduct electricity. This happens because the gas becomes ionized: radiation knocks out one or more electrons (negative particles), leaving positive charges behind. Rutherford also discovered that uranium emitted **two types of radiation, which he named alpha and beta**. His alpha rays were later identified as particles that are the nuclei of helium atoms, while beta rays were found to be streams of electrons. Both were the by-products of radioactive decay.

In March, the Imperial Patent Office in Berlin **trademarked a**

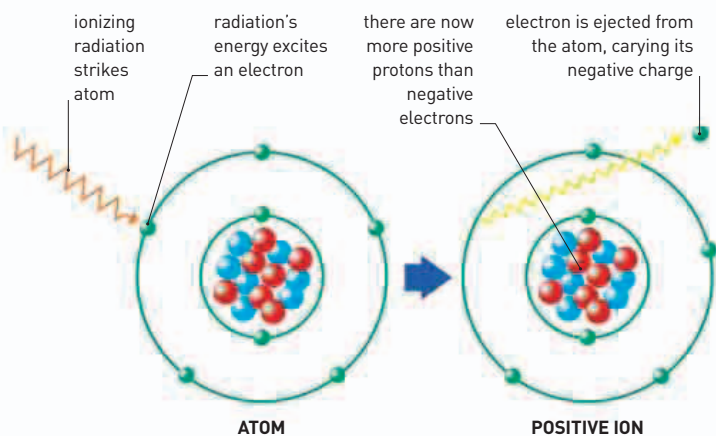
45,000 TONS

THE QUANTITY OF **ASPIRIN** CONSUMED GLOBALLY EACH YEAR

new drug for Bayer, a German pharmaceutical company. The drug was **aspirin**, a painkiller that had been developed by Bayer's scientists two years earlier. Aspirin would become **the world's best-selling drug**.

Effect of ionization

Ionizing radiation (alpha and beta particles, gamma rays, and X-rays) carries enough energy to create ions (charged particles) from atoms.



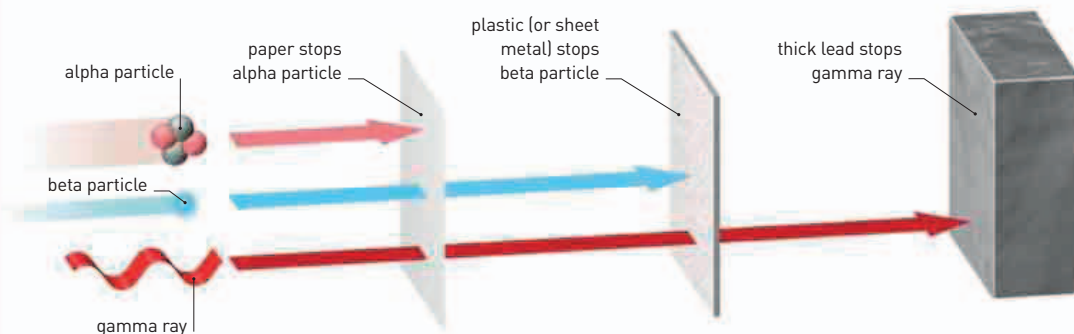
1900

420 FEET

THE LENGTH OF THE **FIRST ZEPPELIN**



Zeppelin LZ1 had its maiden flight over southern Germany in 1900.



THE FRENCH CHEMIST PAUL VILLARD (1860–1934) announced that he had found a **third type of radiation** only a year after the discovery of alpha- and beta-radiation. Villard's rays, emitted by radium salts, were far more penetrating: they were similar to X-rays but had shorter wavelengths and high energy. **Rutherford later called them gamma rays.**

July saw the **first flight of the rigid airship** named after the German **Count Ferdinand von Zeppelin** (1838–1917). Its light-alloy framework—buoyed by an internal system of hydrogen balloons—proved difficult to control. It hailed the start of a period of commercial airship success. The program was scrapped after the fatal Hindenburg crash of 1937.

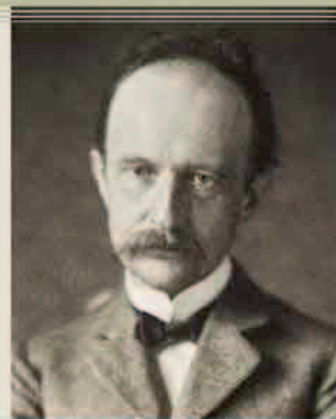
Penetration of radiation

Alpha particles cannot penetrate paper, unlike smaller beta particles. Gamma radiation is not made up of particles and its high-energy rays are stopped only by lead.

In October, German physicist Max Planck had a **theory that proposed a new way of looking at physics**. He was interested in the science behind an everyday phenomenon—darker objects are warmed more by light than paler ones. The theoretically darkest object, a so-called black body, absorbs all electromagnetic radiation, including visible light—and then is a perfect emitter of this radiation. Planck reasoned that there were discrete vibrations of atoms in a body, equivalent to "packages" of energy—which when added together give the total amount of energy emitted. The idea that radiation, such as light, comes in packages of energy later called quanta was **the foundation of quantum physics**.

MAX PLANCK (1858–1947)

Physicist Max Planck studied in Munich and Berlin, and became a professor at Kiel, then Berlin. He helped organize the first **Solvay Conference for Physics** in 1911, when scientists met to discuss quantum theory. He was awarded the Nobel Prize in 1918. Unlike many scientists, Planck remained in Germany during the Nazi government.



January Ernest Rutherford describes his discovery of **alpha and beta radiation**

March 6 German Patent Office registers **aspirin** as a brand name for acetylsalicylic acid for Bayer AG

February American **Henry Osborne** describes **adaptive radiation**: plants and animals can evolve into several species by adapting to different ecological niches

November 6 **Pierre and Marie Curie** describe a radioactive gas given off by radium, later named **radon**

January Ernest Rutherford first describes the idea of emission of radioactivity in terms of half-life

March 23 **Karl Landsteiner** notes a reaction when **serum and blood** are mixed: the basis of blood group theory

March 26 **Hugo de Vries** explains inheritance experiments and later credits Gregor Mendel with formulating laws of heredity

April 9 **Paul Villard** describes his discovery of the radiation later known as **gamma rays**

May US Army physician **Walter Reed** begins a study that later confirms the yellow fever virus is carried by mosquitoes

June Japanese chemist **Jokichi Takamine** and **Keizo Uenaka** crystallize epinephrine: the first hormone to be isolated

July 2 Zeppelin LZ1, the first **successful rigid airship**, makes its maiden flight over Lake Constance, Germany